

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
INSTRUCTION DIVISION
FIRST SEMESTER 2026-2027
Course Handout Part II

Date: 01/08/2026

In addition to part-I (General Handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

Course No. : BITS F464 / CS U311
Course Title : Machine Learning
Instructor-in-charge : Rakhi Agrawal (rakhi.agrawal@pilani.bits-pilani.ac.in)
Lab. Instructor : Priyanka Jain (p20240498@pilani.bits-pilani.ac.in)

Catalog Description

Machine Learning is an exciting sub-area of Artificial Intelligence which deals with designing machines which can learn and improve their performance from examples/experience. This course introduces the student to the key algorithms and theory that forms the core of machine learning. The course will cover all major approaches to learning namely, supervised, unsupervised, semi-supervised, and reinforcement learning. The course emphasizes various techniques, which have become feasible with increased computational power and our ability to produce and capture huge volumes of data. The topics covered in the course include regression, decision trees, support vector machines, artificial neural networks, Bayesian techniques, Hidden Markov models etc. Some advanced topics like active, deep, & topological learning will also be covered.

Text Books:

Tom M. Mitchell, Machine Learning, The McGraw-Hill Companies, Inc. International Edition 1997.

Reference Books:

1. Christopher M. Bishop, Pattern Recognition & Machine Learning, Springer, 2006.
2. Marsland Stephen, Machine Learning – An Algorithmic Perspective, 2e, CRC Press, 2015.
3. Alpaydin Ethem. Introduction to Machine Learning, 3e, PHI, 2014.
4. Andrew Ng, Machine Learning Yearning, Draft Version.
5. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, Deep Learning, MIT Press, 2016.

PREREQUISITES*

1. Probability theory 2. Linear Algebra 3. Optimization & Search	Self Study	Ch.2, Appendix C – R1
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** A good understanding of these topics is essential*

LECTURE PLAN

Topic	Topic Details	# Lectures	Chapter Reference
Overview	Introduction to Machine Learning 1. Course Motivation/Objectives 2. Recent ML success stories 3. Machine learning & function approximation 4. Early introduction to Neural Networks	3	Ch. 1 - TB Ch.1 – R1,R2,R3 Ch. 5 – R5
Types of Learnings	Quick introduction to different types of Learnings: 1. Supervised 2. Unsupervised 3. Semi-supervised 4. Active Learning 5. Reinforcement Learning	1	Ch.1 – R2, R3 + Class Notes

High-dimensional data & Curse of Dimensionality	Characteristics of High-dimensional data, Curse of Dimensionality (CoD) problems & dimensionality reduction techniques (PCA/SVD)	2	Ch.1 – R1
Frequentist vs. Bayesian Approach to Probability	1. Parameter estimation through coin toss experiment 2. Beliefs and Probability 3. Likelihood vs. Posteriors 4. MLE vs. MAP 5. Bayesian Inference Engine	1	Ch. 6 – TB Ch. 16 – R3
Supervised Learning	Regression 1. ML concepts through Polynomial Curve Fitting a. Model flexibility & selection b. Overfitting/underfitting c. Regularization d. Bias-Variance Decomposition 2. Linear basis function models 3. Bayesian Linear Regression	3	Ch. 3 – R1
	Classification 1. Discriminant Functions (two class and multiclass) a. Least Square b. Fisher's Linear Discriminant (FLD) c. Perceptron Algorithm 2. Probabilistic Generative Classifiers (NBC) 3. Probabilistic Discriminative Classifiers (LR) 4. Decision Tree Learning 5. Artificial Neural Networks a. Activation Functions b. Backpropagation Algorithm c. Gradient Descent & variants 6. Support Vector Machines a. Maximal margin hyperplane b. Structural risk minimization c. VC dimension/Shattering d. Soft-margin SVM e. Non-linear SVM i. Kernel Methods ii. Kernel Trick 7. Ensemble Learning a. Bagging b. Boosting (Adaboost) c. Random Forests	14	Ch. 4 – R1 Ch. 10 – R3 Ch. 4 – R1 Ch. 4 – R1 Ch. 3 - TB Ch. 12 – R2 Ch. 9 – R3 Ch. 4 – TB Ch. 5 – R1 Chs. 3,4 – R2 Ch. 11 – R3 Ch. 13 – R2 Ch. 17 – R3 Ch. 8 – R2 Ch. 13 – R3
Unsupervised Learning	Clustering 1. K-means & Kernel K-means 2. Mixture of Gaussians 3. Expectation Maximization (EM) Clustering 4. Self-organizing Maps (SOM)	5	Ch. 9 – R1 Ch. 14 – R2 Ch. 7 – R3
Reinforcement Learning	1. Exploration vs. Exploitation tradeoff 2. Action and state spaces 3. Action Selection: Greedy, e-greedy, & Softmax 4. Markov Decision Process (MDP) 5. Q-Learning & Sarsa Algorithms 6. Case Study: AlphaGo Zero	5	Ch. 13 – TB Ch. 11 – R2 Ch. 18 – R3

Active Learning	1. Learning threshold & interval functions 2. Active Learning Scenarios a. Membership query synthesis b. Pool-based sampling c. Stream-based sampling 3. Query Strategy Frameworks a. Uncertainty sampling (Least confident, Margin sampling, & Entropy) b. Query by committee (QBC) & Version spaces (Vote entropy KL divergence) 4. Case study – Medical image analytics	4	Class Notes
Deep Learning	1. Introduction to Deep Learning 2. n-D Convolution, CNN, & 3D-CNN 3. RNN & LSTM	3	R5
Advanced Topics: Federated Learning & Edge AI	1. Privacy-preserving ML: Federated Learning a. Model communication & aggregation (sketched & structured updates) b. Hierarchical Federated Learning c. Federated Clustering 2. Edge AI	3 Total Lectures: 44	Class notes & research articles

Evaluation Scheme:

Component	Duration	Weightage	Date (Time)
Midsem (closed book)	90 Mins.	30%	10/10 AN1
Quizzes (02) (closed book)	45 Mins.	20%	25/09, 13/11 - during lecture slot
Assignments (02) (non-evaluatory)*	Take Home	–	–
Weekly labs (open book)	2 hrs each	10%	Weekly
Comprehensive Exam (partly open book)	3 Hours	40%	16/12 FN

* Questions based on assignments will be asked in Midsem (10% - closed book) & Comprehensive exams (10% - open book). Students who wish to get input and/or feedback on assignments can request an appointment or come during chamber consultation hours.

Labs in Python: Supervised 2-hour labs will be conducted every week as per the timetable. The best 10 labs will be considered for evaluation.

Notices: All notices will be displayed on NALANDA only.

Chamber Consultation Hour: F 4.00 to 5:00 pm (6121-X, NAB). Interested student(s) must inform the instructor by email beforehand if consultation is required.

Makeup Policy (Midsem & Comprehensive Exam): To be granted only in case of serious illness or emergency. **NO makeup for quizzes.**

Plagiarism Policy: If any student is found to be involved in any kind of plagiarism in any of the evaluation components, the matter will be directly reported to the Examination Committee.

NC Policy: Students securing 10% or less marks will get an NC grade. Students in the [10-15%] bracket are also likely to get NC.

Instructor-in-charge
BITS F464